

Topic

A Step towards Water Neutrality by the Cement Sector

Date

In 2018-19, mega Indian cities such as Chennai and Bangalore highlighted the severity of water crises in the country. In 2019, 330 million people in the country were affected by drought. As per the Composite Water Management Index (CWMI) report released by the NITI Aayog in 2018, 21 major cities, including Delhi, Bengaluru, Chennai and Hyderabad, would be hitting zero groundwater levels by 2020 and nearly 100 million people would be affected. The CWMI report also stated that by 2030, the country's water demand would be twice the volume of available supply those are already living in the "Day Zero" scenario.

In the past few decades there has been a massive increase in groundwater extraction where India ranks one, pumping out 25 per cent of the world's groundwater reserves. Water neutrality is when you preserve more water than you consume. Cement factories which are spread across various geographical locations face water issues not only for their operations but also for the communities living around their plants. For any water intensive Cement Company, a water neutral programme can not be successful until the initiatives are taken beyond the boundaries of the cement plants. The cement sector, which until a few years ago was notorious for guzzling millions of gallons of water and spewing pollutants into the environment, has turned the situation around due to the advancement of the technology used. Previously, cement manufacturing

was based on wet process, but now it has switched over to dry process. This has reduced the water footprint of cement factories considerably, limiting their consumption for captive power plant operations ranging from 16MW to 45MW. Such plants were put up earlier by companies to meet their energy requirements. However, cement manufacturers are now opting for heat recovery plants, which are more suitable and cost effective. The second major requirement of water is from the residential colonies in and around cement factories for domestic use. Third requirement is for dust suppression, dust quenching and horticulture. Dalmia Cement, which is currently three times water positive, has adopted the process almost half a decade ago. This leading company has set suitable targets for the next decade. Water neutrality programmes including water storage, rainwater harvesting, groundwater level recharging, recycling of waste water, availability of potable water to the communities, setting up of check dams and water storage facilities. Most of these initiatives are done with the involvement of community and local body officials. Ashwani Pahuja, Chief Sustainability Officer, Dalmia Cement (Bharat), points out, "Initially there was resistance from the locals until they were made aware of the benefits. Now the participation from the local communities is much higher than when the programme was initiated. For water recycling, the challenge was the availability of technologies in the country, particularly air-cooled condensers etc.

As a first step, most of the Cement Companies harvested water in the rehabilitated mining pits and cleaned it in the demineralisation (DM) plant before supplying as portable water. The water which is cycled through DM Plant and RO Plant goes through a boiler. Similarly, waste water is used for dust suppression and horticultural activities. Also, water from the factory colonies is treated and used for gardening and farming. The project included creation of check-dams on the existing rivers, digging of water harvesting pits, interlinking of canals and thereby creation of a sweet water buffer. The result was very encouraging and positive. The Saline line which had advanced about 15km inwardly was pushed back by out 8km. Over the last 19 years, 20 to 25 villages spread along 65 to 70km of coastline have been covered under the project.

Besides water conservation and water recycling programmes, the Cement factories also devise new and innovative ways to reduce water consumption during production of cement. The target of its sustainable goal is to become at least 20 times water positive by 2025, which means the company aims to achieve four times in the next five years. This still requires the additional implementation of water prudent potential of 31 million m^3 . It ~~is~~ also means increasing rainwater harvesting, increasing water bodies, recycling of water and much more. Most of the Cement factories are also looking at more sustainable ways to

contribute to the fight against climate change and adopting renewable energy, implementing energy-efficiency programmes etc. Appreciable are the initiatives taken by the Cement sector, which is always criticised for huge water spend and pollution. If the Cement sector can contribute to fight against water scarcity, other industries, too, can take a cue and embrace water saving and water conservation technologies.

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